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# PAPERS OVER TIME

UNFINISHED LECTURE NOTES

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# *Contents*

|                                                |   |
|------------------------------------------------|---|
| <i>Zebrafish Phenotype Pattern Recognition</i> | 7 |
|------------------------------------------------|---|



## *Introduction*

PAPERS OVER TIME reads the field through its primary literature. Each session lifts a single paper, situates it in its moment, and asks what it changed and what it left unfinished. The aim is fluency in the genre and taste in the questions.



# Zebrafish Phenotype Pattern Recognition

2026-05-08 · local draft

## COUNTING EMBRYOS

A SINGLE SCREEN produces thousands of embryos. Each one has to be inspected at fixed time points to decide which ones carry the phenotype of interest. By hand, the screen never finishes.

IMAGING HAD OUTPACED SCORING. Embryos could be photographed faster than a biologist could classify them. The bottleneck had moved.

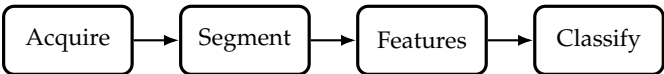
AUTOMATED PHENOTYPE PATTERN  
RECOGNITION OF ZEBRAFISH FOR  
HIGH-THROUGHPUT SCREENING  
Schutera et al. (2016)  
*Bioengineered* 7(4)



|          |                                                              |
|----------|--------------------------------------------------------------|
| Question | Can a classifier replace the human at the microscope?        |
| Setting  | Four phenotypes at 72 hpf, reduced to wild type vs. variant. |
| Headline | 79% to 99% accuracy, no user interaction.                    |

BACHELOR THESIS.  
The first paper, written before any of this was supposed to be a career.

MODEL ORGANISM. Zebrafish (*Danio rerio*) embryos are transparent, cheap, and develop fast. The phenotype is visible at 72 hours post fertilization.



NO DEEP LEARNING. The data set is small, the phenotypes are well described by intensity and geometry, and a domain expert can name the features that matter. Classical pattern recognition ships.

THE CONTRIBUTION IS THE LOOP. Rig plus software runs unattended on a real screening question and returns numbers a biologist can act on. Algorithm second, system first.

THE MOVE. Treat acquisition and classification as one system. The features the classifier needs are the features the rig is set up to capture.

TAKE IT FROM HERE. Drift across screens, rigs, and batches becomes the dominant problem. Labels become the next bottleneck. The later papers pick the thread up.